

Member ID: _____

Time: _____

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DEVICE CONFIGURATION AND TROUBLESHOOTING (305)

NATIONAL 2023

Total Points (50 @ 2 points each) _____ (100 points)

Test Time: 60 minutes

GENERAL GUIDELINES:

Failure to adhere to any of the following rules will result in disqualification:

1. Member must hand in this test booklet and all printouts if any. Failure to do so will result in disqualification.
2. No equipment, supplies, or materials other than those specified for this event are allowed in the testing area. No previous BPA tests and/or sample tests (handwritten, photocopied, or keyed) are allowed in the testing area.
3. Electronic devices will be monitored according to ACT standards.

Multiple Choice Questions

Directions: Identify the letter of the choice that best completes the statement or answers the question.

1. Which of the following is the smallest motherboard form factor?
 - A. Micro-ATX
 - B. ATX
 - C. Mini-ITX
 - D. Nano-ITX
2. How many PCI/PCIe slots do mini-ITX motherboards normally have?
 - A. 3
 - B. 2
 - C. 1
 - D. None
3. Which of the following statements is **not** true?
 - A. ExpressCard/54 card can fit into an ExpressCard/34 slot
 - B. ExpressCard/34 can fit into an ExpressCard/54 slot
 - C. Express cards require drivers to function
 - D. ExpressCard are hot-swappable
4. Compatibility mode is a group of Windows settings that allow _____ applications and drivers to work in a(n) _____ operating system even though they are not compatible.
 - A. older, newer
 - B. newer, older
 - C. Windows, Linux
 - D. Linux, Windows
5. Pressing **Win+V** will _____.
 - A. cut selected text out of a document and paste it to the clipboard
 - B. close the active window
 - C. close all open windows
 - D. open clipboard history
6. Which two services does Windows provide for an ExpressCard?
 - A. A bus service and a spool service
 - B. A storage service and a network service
 - C. An express service and a card service
 - D. A socket service and a card service
7. A hard drive can have more than one volume, each with its own root directory.
 - A. True
 - B. False

8. These steps are performed inside the image drum cartridge on newer laser printers:
 - A. Charging, transferring, fusing
 - B. Developing, transferring, fusing
 - C. Cleaning, charging, exposing
 - D. Charging, developing, processing
9. What types of printers can support duplexing?
 - A. Only laser printers
 - B. Some laser printers and inkjet printers
 - C. Some laser printers and thermal printers
 - D. Some laser printers, inkjet printers and impact printers
10. Which partitioning method has a maximum capacity of 2 TB?
 - A. MBR
 - B. FAT 32
 - C. GPT
 - D. NTFS
11. What is the latest Windows OS that can be installed on a computer that has a 1.2 GHz processor with 32-bit and 64-bit compatibility, 1 GB RAM and 32GB hard drive?
 - A. Windows 8, 32-bit OS
 - B. Windows 8, 64-bit OS
 - C. Windows 10, 32-bit OS
 - D. Windows 10, 64-bit OS
12. Which of the following ports does **not** provide both an audio and digital video signal?
 - A. S-Video
 - B. Composite Video
 - C. DVI
 - D. All of the other answers are correct
13. What should you do first if you notice smoke coming from the computer case?
 - A. Disconnect the power from the computer
 - B. Take a picture
 - C. Find a class C fire extinguisher rated for electrical fires
 - D. Increase ventilation to “air out” the computer
14. Which Wi-Fi standard has the potential to provide the fastest data transfer speed and supports the 2.4-GHz radio frequency?
 - A. IEEE 802.11b
 - B. IEEE 802.11g
 - C. IEEE 802.11ac
 - D. IEEE 802.11n

15. Most current external hard drives and optical drives use a(n) _____ interface.
- A. IDE
 - B. USB
 - C. SCSI
 - D. SATA
16. “Overclocking failed” error appears but overclocking is not being used. What should be checked next?
- A. Memory diagnostics
 - B. Power supply
 - C. Hard drive
 - D. CMOS battery
17. Which of the following has a PC rating of PC4-19200?
- A. DDR4-1866
 - B. DDR4-2133
 - C. DDR4-2400
 - D. DDR4-2666
18. Which combination of memory modules would allow the architecture to operate in a quad-channel mode?
- A. 2 x DDR2 modules
 - B. 4 x DDR2 modules
 - C. 2 x DDR4 modules
 - D. 4 x DDR4 modules
19. Which type(s) of memory cache are located on the processor die?
- A. L1
 - B. L2
 - C. L3
 - D. L1 and L2
20. 32-bit or 64-bit Windows OS can be installed on _____ processors.
- A. x86
 - B. x86-64
 - C. IA-64
 - D. All of the other answers are correct
21. Which of the following would be a valid step in installing a new processor chip?
- A. apply a thin layer of thermal compound to the top of the processor
 - B. align the processor notches with the socket posts or line up the marked corners
 - C. wear an ESD strap or anti-static gloves
 - D. All of the other answers are correct

22. A product key is required for both Professional and Enterprise installations of Windows?
- A. True
 - B. False
23. Which of the following is the CIDR notation for this subnet mask: 255.255.255.0?
- A. /8
 - B. /16
 - C. /20
 - D. /24
24. The main function of the system clock is to ____.
- A. synchronize all activities of the motherboard
 - B. keep the current date and time
 - C. keep track of how many cycles are required by the CPU to run a process
 - D. monitor latency of deferred procedure call (DPC) requests
25. What is the first thing you should do when troubleshooting mobile device hardware?
- A. Open the device case carefully and visually inspect
 - B. Review the warranty information
 - C. Download the service manual
 - D. Consider the cost of repair versus replacing the device
26. Which of the following is a standard rotation speed for a magnetic hard drive?
- A. 5,200 RPM
 - B. 7,400 RPM
 - C. 10,000 RPM
 - D. None of the other answers are correct
27. _____ is a Windows 8 and 10 utility used to back up user data from several locations to another media and can maintain multiple versions of files.
- A. Version Manager
 - B. File History
 - C. Backup and Restore
 - D. System Restore
28. Which of the following CMD or PowerShell commands **cannot** be used to create a directory or folder?
- A. Create-Folder -Path
 - B. mkdir
 - C. New-Item -ItemType Directory
 - D. md
29. Which of the following tasks can be performed through Disk Management?
- A. Initialize a drive
 - B. Shrink a volume
 - C. Change a drive letter
 - D. All of the other answers are correct

30. Which of the following command line utilities can be used to perform the same functions as Disk Management?
- A. devmgmt.msc
 - B. fsutil
 - C. diskpart
 - D. chkdsk
31. The IPV6 address FE80:0000:0000:0000:C040:0000:00B6:F100 can also be written:
- A. FE80::C040::B6:F100
 - B. FE80::C040:0000:B6:F1
 - C. FE80:::C040::00B6:F100
 - D. FE80::C040:0:B6:F100
32. You are troubleshooting a network problem on a Linux system. What command is needed to view the IP address of the system?
- A. ipconfig
 - B. ping
 - C. netsh
 - D. ifconfig
33. Which of the following could be the MAC address of a device?
- A. C0-00-D9-FE-4A-11
 - B. D1-3A-00-01-42-7E-16-B0-D0-79-24-19-FA-73
 - C. 192.168.30.7
 - D. G9-10-3C-HA-81-23
34. Which network adapter setting in Windows provides the best possible speed and duplex?
- A. Optimized
 - B. Auto Negotiation
 - C. 10 Mbps Full Duplex
 - D. 100 Mbps Half Duplex
35. What is the speed of gigabit ethernet?
- A. 1 GBps
 - B. 1 MBps
 - C. 1000 Mbps
 - D. 1000 Gbps
36. The _____ senses how a user moves a mobile device and can adjust the screen orientation to portrait or landscape.
- A. gyroscope
 - B. accelerometer
 - C. GPS receiver
 - D. multimeter

37. What feature of a network adapter controls which application has network priority?
- A. Large Send Offload
 - B. Quality of Service
 - C. TCP Checksum Offload
 - D. Wait for Link
38. Which of the following secures a hard drive by encrypting the data on it so that it can't be accessed if the hard drive is stolen and installed in another machine?
- A. TPM chip and BitLocker Encryption
 - B. Hard Drive Password Protection
 - C. Deduplication
 - D. Software Defined Storage
39. A computer technician would probably want to _____ a Windows computer before donating it or recycling to remove all user data and settings.
- A. refresh
 - B. reset
 - C. reinstall
 - D. recover
40. What is the minimum number of disks needed to implement RAID 5?
- A. 1
 - B. 2
 - C. 3
 - D. 4
41. What is the chief function of spacers or standoffs?
- A. Accommodate a smaller hard drive in a wider bay
 - B. Separate expansion cards to prevent overheating
 - C. Prevent motherboard short-circuits when the system is running
 - D. Increase airflow around the exterior of the case and absorb vibrations
42. Where is the drive bay typically located in a tower or desktop computer case?
- A. Front
 - B. Back
 - C. Side
 - D. None of the other answers are correct
43. _____ are examples of front panel connectors
- A. POWER SW, HDD LED, SPEAKER
 - B. USB port, audio jacks
 - C. RJ45, HDMI, eSATA
 - D. None of the other answers are correct

44. Which of the following specifications are important to consider when purchasing a gaming monitor?
- A. Resolution
 - B. Refresh rate
 - C. Response time
 - D. All of the other answers are correct
45. Which of the following is a valid loopback address?
- A. 127.0.0.1
 - B. 255.255.255.0
 - C. 127.0.0.254
 - D. 255.255.255.254
46. Which Windows command(s) could be used to repair a network connection on a computer with no connectivity after verifying that network hardware is functional, and cables are properly connected?
- A. ipconfig /all
 - B. ipconfig /release
 - C. ipconfig /registerdns
 - D. ipconfig /renew
47. Fault tolerance is ____.
- A. The ability of a memory module to detect and correct an error in a single bit
 - B. The degree of failure a system can handle without data loss
 - C. The capacity of an AI to accept when it is wrong
 - D. Amount of vibration an optical drive can handle without the media “skipping”
48. When building a network, a/an ____ cable is used to connect two devices that are alike, such as two computers or two hubs.
- A. Patch
 - B. Cross-over
 - C. Straight-through
 - D. RJ-11
49. The ____ protocol is used to securely control a remote computer using encrypted transmissions.
- A. SMB2
 - B. SFTP
 - C. Telnet
 - D. SSH
50. The ____ command is used to restore critical Windows system files.
- A. restore
 - B. chkdsk /r
 - C. sys /repair
 - D. sfc /scannow